Number of Contiguous Superregular $3 \times n$ Matrices in $\mathbb{F}_q$

A matrix is contiguous superregular if, for each possible size k, every $k \times k$ submatrix with consecutive rows and columns extracted from the orginal matrix has nonzero determinant. Denote $C_n(q)$ the number of such matrices over $\mathbb{F}_q$. All notations follow the convention of Sanna.

Theorem 1. *For every prime power q and every $n \geq 2$, define*

$$N_n(q) = \frac{C_n(q)}{(q-1)^{n+2}}$$

Then

$$N_{n+2}(q) = (q-3)^2 N_{n+1}(q) + (q-2)^2(q-3)N_n(q) \tag{1}$$

with initial values

$$N_2(q) = (q-2)^2, \qquad N_3(q) = (q-2)(q-3)(q^2-4q+5) \tag{2}$$

Equivalently,

$$C_{n+2}(q) = (q-1)(q-3)^2 C_{n+1}(q) + (q-1)^2(q-2)^2(q-3)C_n(q) \tag{3}$$

1 Normalisation

For a matrix $A = (a_{ij})$, the CSR condition implies $a_{ij} \neq 0$ for all i, j, so every division used later is defined.

Lemma 2. *Let $A = (a_{ij})$ be an $m \times n$ CSR matrix. Define a pivot position (r, c). Define*

$$b_{ij} = \frac{a_{rc}a_{ij}}{a_{ic}a_{rj}} \tag{4}$$

Then $B = (b_{ij})$ is CSR, and its rth row and cth column are ones, and every normalised CSR matrix B of this form corresponds to $(q-1)^{m+n-1}$ CSR matrices A.

Proof. Substituting $i = r$ or $j = c$ gives

$$b_{rj} = 1, \qquad b_{ic} = 1$$

So B has rth row and cth column of ones.

The transformation from A to B can be performed by multiplying row i by a_{ic}^{-1} and then multiplying column j by $a_{rc}a_{rj}^{-1}$. So it doesn't change the zero / nonzero status of the determinant. It follows that A is CSR if and only if B is CSR.

Now we recover all matrices A from a fixed normalised B. Fix a row and column of B to be ones, and choose arbitrary elements

$$u_i \in \mathbb{F}_q^\times \quad (1 \leq i \leq m), \qquad v_j \in \mathbb{F}_q^\times \quad (j \neq c)$$

and $v_c = u_r$. Then

$$a_{ij} = \frac{u_i v_j}{u_r} b_{ij} \tag{5}$$

Since $b_{ic} = b_{rj} = 1$,

$$a_{ic} = u_i, \qquad a_{rj} = v_j$$

It suffices to show that this transformation is the exact inverse operation of equation (4). From equation (5), the reconstructed matrix is CSR when B is CSR.

There are m freely chosen elements u_i and $n-1$ freely chosen elements v_j with $j \neq c$. Hence there are $m+n-1$ independent nonzero choices and therefore $(q-1)^{m+n-1}$ choices. $\square$

2 The $3 \times n$ case

For a $3 \times n$ matrix, choose the middle row and the first column as the pivot. Every normalised matrix then has the form

$$B = \begin{pmatrix} 1 & x_1 & x_2 & \cdots & x_{n-1} \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & y_1 & y_2 & \cdots & y_{n-1} \end{pmatrix}, \qquad x_0 = y_0 = 1 \tag{6}$$

Let $N_n(q)$ be the number of normalised CSR matrices of the form equation (6). By lemma 2,

$$C_n(q) = (q-1)^{n+2} N_n(q) \tag{7}$$

We can now only consider the CSR condition for the normalised matrix. The 1×1 determinants imply

$$x_j \neq 0, \qquad y_j \neq 0 \qquad (1 \leq j < n) \tag{8}$$

For 2×2, we have

$$\det \begin{pmatrix} x_j & x_{j+1} \\ 1 & 1 \end{pmatrix} = x_j - x_{j+1}$$

and

$$\det \begin{pmatrix} 1 & 1 \\ y_j & y_{j+1} \end{pmatrix} = y_{j+1} - y_j$$

The 2×2 conditions are therefore

$$x_j \neq x_{j+1}, \qquad y_j \neq y_{j+1} \qquad (0 \leq j < n-1) \tag{9}$$

For 3×3, we need to consider only one case

$$\det \begin{pmatrix} x_j & x_{j+1} & x_{j+2} \\ 1 & 1 & 1 \\ y_j & y_{j+1} & y_{j+2} \end{pmatrix} \tag{10}$$

3 Ratio States

For $1 \leq j < n$, define

$$a_j = \frac{x_j}{x_{j-1}}, \qquad b_j = \frac{y_j}{y_{j-1}}, \qquad D = \mathbb{F}_q^\times \setminus \{1\} \tag{11}$$

The divisions are defined by equation (8). Moreover,

$$x_j \neq x_{j-1} \iff a_j \neq 1, \qquad y_j \neq y_{j-1} \iff b_j \neq 1$$

It follows that the 1×1 and 2×2 conditions together are equivalent to

$$(a_j, b_j) \in D^2 \qquad (1 \leq j < n) \tag{12}$$

From equation (11), there is

$$x_j = \prod_{i=1}^{j} a_i, \qquad y_j = \prod_{i=1}^{j} b_i \tag{13}$$

because $x_0 = y_0 = 1$. Consider two consecutive ratio states

$$(a, b) = (a_j, b_j), \qquad (c, d) = (a_{j+1}, b_{j+1})$$

They describe the three consecutive columns $j-1, j, j+1$. From the definitions,

$$x_{j-1} = \frac{x_j}{a}, \qquad x_{j+1} = cx_j$$

and

$$y_{j-1} = \frac{y_j}{b}, \qquad y_{j+1} = dy_j$$

The general 3×3 determinant case is therefore

$$\det \begin{pmatrix} \frac{x_j}{a} & x_j & cx_j \\ 1 & 1 & 1 \\ \frac{y_j}{b} & y_j & dy_j \end{pmatrix}$$

Divide the first row by x_j and third row by y_j (which also doesn't change the zero / nonzero status). We therefore have

$$\det \begin{pmatrix} \frac{1}{a} & 1 & c \\ 1 & 1 & 1 \\ \frac{1}{b} & 1 & d \end{pmatrix} = (\frac{1}{a} - 1)(d-1) - (c-1)(\frac{1}{b} - 1) \tag{14}$$

If the determinant is zero (meaning B is not CSR), we have

$$\frac{d-1}{c-1} = \frac{\frac{1}{b} - 1}{\frac{1}{a} - 1}$$

For $(\alpha, \beta) \in D^2$, define

$$f(\alpha, \beta) = \frac{\frac{1}{\beta} - 1}{\frac{1}{\alpha} - 1} = \frac{\alpha(\beta - 1)}{\beta(\alpha - 1)}, \qquad g(\alpha, \beta) = \frac{\beta - 1}{\alpha - 1} \tag{15}$$

So B is not CSR if and only if

$$g(c, d) = f(a, b)$$

Consequently, the matrix is CSR when

$$f(a, b) \neq g(c, d) \tag{16}$$

If we write a normalised $3 \times n$ matrix as a sequence

$$(a_1, b_1), \ldots, (a_{n-1}, b_{n-1}) \in D^2$$

such that

$$f(a_j, b_j) \neq g(a_{j+1}, b_{j+1}) \qquad (1 \leq j < n-1)$$

then we call the sequence legal, and the matrix is CSR.

4 Counting

For $u, v \in \mathbb{F}_q^\times$, define

$$H_{u,v} = \text{number of } (a,b) \in D^2 \text{ satisfying } g(a,b) = u,\ f(a,b) = v \tag{17}$$

Lemma 3. *For every finite field,*

$$H_{u,v} = \begin{cases} q-2, & u = v = 1, \\ 1, & u \neq 1,\ v \neq 1,\ u \neq v, \\ 0, & \textit{others} \end{cases} \tag{18}$$

Proof. Suppose first that $u = 1$. Then

$$g(a,b) = \frac{b-1}{a-1} = 1$$

which implies $b = a$, and substitution into equation (15) gives $v = f(a,b) = 1$. Conversely, $f(a,b) = 1$ implies

$$a(b-1) = b(a-1)$$

and hence $a = b$, which gives $g(a,b) = 1$. Thus there is no solution when one of u, v equals 1. They have to both equal 1. If $u = v = 1$, the states are (a,a) for $a \in D$. Since $|D| = q-2$, this gives $H_{1,1} = q-2$.

Now consider $u, v \neq 1$. The equation $g(a,b) = u$ gives

$$b - 1 = u(a-1), \qquad b = ua + 1 - u \tag{19}$$

Using $b - 1 = u(a-1)$ in $f(a,b) = v$ gives

$$v = \frac{au}{b}, \qquad vb = ua$$

Substitution gives

$$v(ua + 1 - u) = ua$$

Expanding gives

$$ua(v-1) = v(u-1)$$

Since $u \neq 0$ and $v \neq 1$, there is the unique solution

$$a = \frac{v(u-1)}{u(v-1)} \tag{20}$$

The equation $vb = ua$ then gives

$$b = \frac{u-1}{v-1} \tag{21}$$

Both values are nonzero. Moreover,

$$a = 1 \iff u = v, \qquad b = 1 \iff u = v$$

Thus the unique solution lies in D^2 exactly when $u \neq v$. □

Assume first that $q \geq 3$, and write

$$w = q - 2, \qquad E = \mathbb{F}_q^\times \setminus \{1\}, \qquad |E| = w \tag{22}$$

For a legal sequence of $L \geq 1$ ratio states, let A_L be the number of sequences whose last state has $f = 1$. For a fixed $v \in E$, let B_L be the number whose last state has $f = v$.

There are $w+1$ possible values that f can take, one of them being 1.

f of last state	number of sequences
1	A_L
v_1	B_L
v_2	B_L
$\vdots$	$\vdots$
v_w	B_L

Hence the total number T_L of legal sequences of length L is

$$T_L = A_L + wB_L \tag{23}$$

For a sequence of length one,

$$A_1 = \sum_u H_{u,1} = w \tag{24}$$

For fixed $v \in E$, u can take any value $E \backslash \{v\}$, so

$$B_1 = w - 1$$

Consider now the length A_{L+1}. If the last state is $f = 1$, then $g = 1$, and there are w distinct states. By equation (16), an old sequence can precede any of these new states when its last f of the old sequence is not 1. The number of such old sequences is

$$T_L - A_L = wB_L$$

Each allowed old sequence can be combined with each of the w new states. Therefore

$$A_{L+1} = w^2 B_L \tag{25}$$

To count B_{L+1}, fix the last state $f = v \in E$. The g for this state can be any $u \in E \setminus \{v\}$, giving $w-1$ possible new states. For any fixed u, the old sequences cannot have last $f = u$. There are B_L such sequences, so the number of allowed predecing sequences is

$$T_L - B_L = A_L + (w-1)B_L$$

So the total count is

$$B_{L+1} = (w-1)(A_L + (w-1)B_L) \tag{26}$$

Equations equations (25) and (26) satisfy

$$\begin{pmatrix} A_{L+1} \\ B_{L+1} \end{pmatrix} = M \begin{pmatrix} A_L \\ B_L \end{pmatrix}, \qquad M = \begin{pmatrix} 0 & w^2 \\ w-1 & (w-1)^2 \end{pmatrix} \tag{27}$$

A normalised $3 \times n$ matrix has columns indexed by $0, 1, \dots, n-1$. There are $n-1$ ratio state for adjacent pairs of columns. Therefore,

$$N_n = T_{n-1} \tag{28}$$

Using equations (23) and (24), we obtain

$$N_2 = T_1 = w + w(w-1) = w^2$$

equation (27) gives

$$A_2 = w^2(w-1) \qquad B_2 = (w-1)(w + (w-1)^2)$$

It follows that

$$\begin{aligned} N_3 = T_2 &= A_2 + wB_2 \\ &= w^2(w-1) + w(w-1)(w+(w-1)^2) \\ &= w(w-1)(w^2+1) \end{aligned}$$

Substituting $w = q-2$ gives

$$N_2 = (q-2)^2$$

and

$$N_3 = (q-2)(q-3)(q^2-4q+5)$$

which are the initial values in equation (2).

5 Recurrence

The trace and determinant of M are

$$\operatorname{tr} M = (w-1)^2, \qquad \det M = -w^2(w-1)$$

Its characteristic polynomial is therefore

$$\lambda^2 - (w-1)^2\lambda - w^2(w-1)$$

By the Cayley–Hamilton theorem,

$$M^2 = (w-1)^2 M + w^2(w-1)I \tag{29}$$

Let

$$V_L = \begin{pmatrix} A_L \\ B_L \end{pmatrix}$$

Since $V_{L+1} = MV_L$, equation (29) gives

$$\begin{aligned} V_{L+2} &= M^2 V_L \\ &= (w-1)^2 MV_L + w^2(w-1)V_L \\ &= (w-1)^2 V_{L+1} + w^2(w-1)V_L \\ \begin{pmatrix} A_{L+2} \\ B_{L+2} \end{pmatrix} &= (w-1)^2 \begin{pmatrix} A_{L+1} \\ B_{L+1} \end{pmatrix} + w^2(w-1) \begin{pmatrix} A_L \\ B_L \end{pmatrix} \end{aligned}$$

Therefore we have

$$A_{L+2} = (w-1)^2 A_{L+1} + w^2(w-1)A_L \tag{30}$$

$$B_{L+2} = (w-1)^2 B_{L+1} + w^2(w-1)B_L \tag{31}$$

Therefore,

$$T_{L+2} = (w-1)^2 T_{L+1} + w^2(w-1)T_L \tag{32}$$

Using $N_n = T_{n-1}$ and $w = q-2$, this becomes

$$N_{n+2} = (q-3)^2 N_{n+1} + (q-2)^2(q-3)N_n$$

which gives equation (1) for $q \geq 3$.

Finally, substitute

$$N_j = \frac{C_j}{(q-1)^{j+2}}$$

into equation (1) and multiply by $(q-1)^{n+4}$. We obtain

$$C_{n+2} = (q-1)(q-3)^2 C_{n+1} + (q-1)^2(q-2)^2(q-3)C_n$$

which proves equation (3) and completes the proof of theorem 1.

6 The $q = 2$ Case

If $q = 2$, then

$$\mathbb{F}_2^\times = \{1\}$$

But then $D = \varnothing$. Every width $n \geq 2$ requires at least one ratio state, so $N_n = 0$ for all $n \geq 2$.